



Toys That Mobilize: Past, Present and Future of Phygital Playful Technology

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Abstract. This exploratory paper focuses on the technological development of the toy medium with an interest in toys' capacity to mobilize *homo ludens*, the playing human. By conducting an extensive literary review on the history and the present of mobile play objects, the study demonstrates how toys have developed from early moving and mechanical automata to playthings that move by themselves through in-built computerized components. The interest is two-fold: By analyzing the historical trajectory of mobile toys, the authors highlight the role of both toys and their players as participants in technologically mediated, phygital play. This hybrid form of playing combines the physicality of playthings with both mechanical and digital features. The results of the review show how toys—character toys in particular—have transformed from entertaining, self-moving spectacles to educational machines that mobilize the player both physically and geographically. Based on the results of the literary review, the authors suggest a continuum that visualizes the development of the types of toys that afford mobility in play of the past, present, and future. The paper concludes with the observation that phygital, playful technologies, such as character toys, have the capacity to influence human well-being in its various dimensions. By making them mobile, toys as a medium invite engagement with phygital playful technologies to enhance physical, cognitive, and social well-being.

Keywords: Toys · Toy mobility · Phygitals · Smart toys · Well-being

1 Introduction: From Mechanical to Phygital Toy Mobility

It has been stated that “the access, the mobility, and the ability to effect change are what will make the future so different from the present” [1]. In this article, a mobile continuum for toys focusing on the past, present, and future is presented, and the ways toys have developed from self-moving automata to become a *mobilizing media* of the 21st century using phygital playful technology have been investigated. Phygital playful technology refers to hybridity in play objects that combines physical playthings with both mechanical and digital affordances. During the past decennia, phygitality has come to be known as a *mobilizing* technology implemented in the design of many new toys.

So far, phygital technologies that combine physical objects and environments with digital and mobile devices and systems have, for example, been studied from the perspectives of games [2, 3], heritage communication [4], and education [5]. In this

paper, the focus is on spontaneous and unstructured play with phygital technologies. The decrease in unstructured playtime can present serious issues for the cognitive, emotional, physical, and social development of children [6]. Again, play with phygital toys can prompt creative play patterns among players. Therefore, the way children play is to some extent influenced by the industry creating new types of play-objects and experiences, such as the toys focused on in this paper, which integrate physical and digital elements known as phygital play-objects [7].

The evolution of playthings enhanced by technologies is inevitable with the invention of new and interesting toys and games being created and released every year. Hybrid, or as referred to in this paper, *phygital* toys and play, challenge a strict dichotomy between physical and digital, non-connected and connected toys by resting on the interaction of material and digital. In other words, traditional, physical toys are merging with digital, “smart” toys [8]. Yelland (2015) has stated that “making new technologies available alongside traditional materials (e.g. blocks) enables and extends playful explorations” [9]. The concept of hybridity, or rather, phygitality, is of essential importance when considering the latest developments around play objects that move by themselves, and even more so that increasingly encourage physical (even geographical) movement of their players. One observation that has guided the interest towards the mobilizing tendency of toys is the insight of the importance of computerized technologies as drivers of human well-being in its various dimensions. Phygital toys represent a growth industry, and they can make a positive contribution to well-being facilitated by toy mobility, discussed in this paper.

There are many types of movement in relation to the physical plaything: mobility in toys may be addressed by considering toys that move by themselves—the historical category of *automata*—or by considering toys as artifacts that are, in some way, made mobile by their players. This exploratory paper sheds light on the development of both of these areas—the toys that are *given* mobility by their designers and manufacturers, and the toys that have the *capacity to mobilize* their users as players. Next, a brief overview of the history of mobile toys is provided.

Automata and Mechanical Toys

Toys as playthings that move by themselves have always intrigued inventors. According to the curators of the *Of Toys and Men* exhibition that was showcased in the early 2010s in European cities, “moving toys [were] developed in the 1800s side-by-side with advances in clockwork construction and the technology of metal plate punching. Lots of cheap mechanical toys were manufactured in Europe in the late 19th century, representing various aspects of life, such as professions, the circus, animals, music and transport. The earliest manufacturers were French and German” [10].

Jaffé stated that “The addition of the simplest mechanism can dramatically transform a static toy into one that moves, introducing new elements of surprise and unpredictability that change and expand the scope for the play activity. This use of mechanics and movements in toy making is not new, and some techniques are almost as old as the first toys themselves” [11, p. 172]. Jaffé further explained that the early historical toys, such as toy animals with wheels or wings from Egypt, Mexico, Russia, India, and Japan, did not only “represent a history of technology and the human fascination with movement, but they also reflect attitudes to the role of God and the meaning of life” [11, p. 173].

In the early days of commercial toys, the first technologically advanced toys came to be known as automata. An *automaton* refers to a self-moving machine, a popular toy-type that predicted the birth of the category of smart toys of today. Technologically-mediated interaction, another affordance of many toys of the present, was already a design feature of playthings of the early 1800s. For example, in 1823, Johann Maelzel, the inventor of the metronome, patented the first talking doll in Paris, which said “mama” and “papa”. Again, during this time in Britain, John Garsad took out a patent for “Automaton Figures and Appliances for Operating the same” for a moving horse and rider [11, p. 179].

Phygital Playful Technology

Toys as an area of playthings have always represented a field of artifacts that have invited artistic and design experiments. For example, it is known that major influencers of their time, da Vinci and Picasso, invented and designed toys. As demonstrated, from the 1800s onwards, commercial toys were produced using wood and metal and sometimes in combination with fabrics, such as dolls and animal figures, and were given movement by mechanical features. In a digitalizing world, it is then of no surprise that the creatives of the industrial toy world have been open to the idea to merge the physicality of toys with the digitality of computerized systems, and later, the connectivity of online environments. Among the earliest digital toys are Mattel Auto Race and the Little Professor, both released in 1976.

The concept of using technology in a way that aligns with the physical world, providing unique hybrid and interactive experiences for the user, has also been referred to as “phygital”. As Pesce already noted in 2000, “the insubstantial world of cyberspace and the world of real objects are beginning to intersect and influence one another” [12, p. 232]. In 2013, *Play it! The Global Toy Magazine* published by the world’s largest toy fair, Nürnberg Spielwarenmesse, stated that “classic and digital play ideas merge to become the toys of tomorrow.” Technological features in playthings, as well as digitalized toys, are believed to be here to stay, and the current mindset seems to support that “iToys do not pose a threat, but should instead be seen as a complement to classical toys” [13]. After this idea was communicated in 2013, hundreds of new phygital toys, combining the physicality of the three-dimensional toy with digitality of computerized systems, have been introduced to the toy market. This paper aims to highlight this development by focusing on one aspect of phygital toys: their mobility.

The research questions are as follows:

- How does playful phygital technology, such as physical toy characters, mobilize players in the 21st century?
- How do connected toys, such as the Internet of Toys (IoToys, like many current coding toys), encourage players to become physically mobile?

This paper is structured as follows. In Sect. 2, the methods of the study are explained and the research material is introduced. Section 3 defines the concept of mobile toys and toys that mobilize their players. In Sect. 4, the results of the study are presented by

demonstrating how it is possible to understand the past and present landscape of toy mobility. Moreover, in this section, a classification of mobilizing toy technologies is proposed by introducing the *Toy Mobility Continuum*. The following section summarizes and discusses the findings of the study. The final section, Sect. 6 concludes the paper and suggests ideas for future research in the area of mobilizing toys.

2 Method

2.1 Research Design

To investigate the past, present, and future of the toys that mobilize players, the authors conducted an extensive literature review on the historical trajectory of toys with mechanically mobilizing affordances as well as playthings with phygitally emerging mobilizing capacities.

In this study, the triangulation approach was used, and mixed methods were applied. Previous empirical study materials [14–19] were used in combination with a literature review to develop a synthesis of mobile toys and toys that mobilize to create a continuum for mobile toys focusing on the past, present, and future.

This section is based on a review of existing literature from articles that are published in reputed journals and conference papers and focus on the mobility toys and toys that mobilize. A data set of references available for or related to the domain of mobile toys and toys that mobilize was obtained through an online search of select databases, such as Emerald, Google Scholar, Research Gate, Academia Edu, and ACM. The keywords used to perform the search included “mobile toys,” “robot toys,” “smart toys,” “Internet of Toys,” and “toys that mobilize,” which were used to obtain a list of references pertaining to literature available on the subject.

It is important to note that the analysis excluded research papers that refer to patents of technologically-enhanced toy designs. Moreover, the analysis did not include descriptions or specifications of toys’ technological affordances that allow movement. Therefore, the aspect of toy mobility has mostly been considered with the player perspective in mind. For current mobile toys to be relevant to the inquiry, references were found in relation to how they have been envisioned to function in play and thus afford movement in relation to the player as either a spectator of the plaything or a co-player that is “set in motion” by the plaything.

A sample of the literature used for the review is provided in Fig. 1.

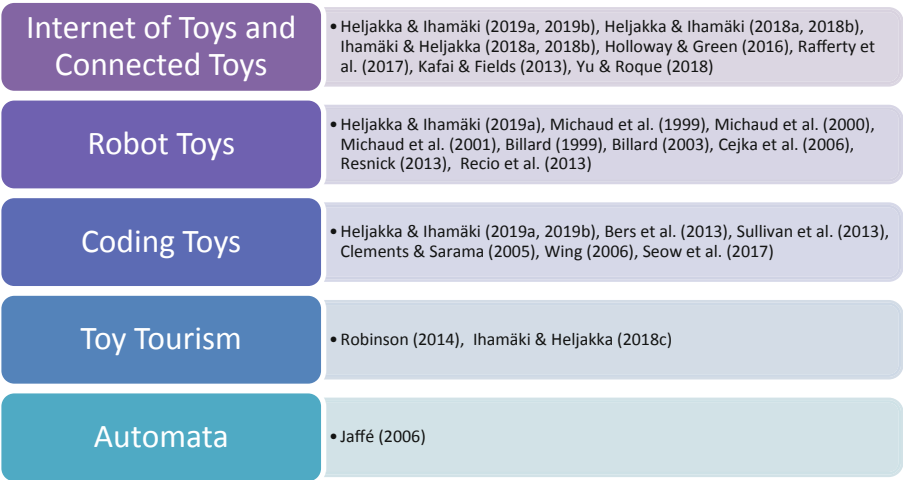


Fig. 1. Literature review sample of toy mobility and toys that mobilize players.

2.2 Convenience Sampling Method: Earlier Empirical Studies

The convenience sampling method was used because accessible literature that was reviewed in combination with a series of empirical studies was selected. These multiple, empirical studies included voluntary teachers and preschool-aged children at a Scandinavian kindergarten who participated in research exploring play patterns related to state-of-the-art phygital toys, such as IoToys and coding toys. This paper presents a synthesis of all the authors’ empirical materials from earlier studies, and the studies that were supported by new research material related to scholarly contributions to the field were extended. Moreover, the newest toy fair-related marketing materials acquired from the toy fairs in Nuremberg and New York from 2019 were employed to understand the potential future of mobile toys and toys that are capable of mobilizing players.

In the case studies conducted by the authors in earlier stages of research, the toy marketers, preschool teachers, and parents of preschool children who participated in playtests with a focus on various toys’ affordances, were analyzed [15, 18]. For example, three types of research materials were employed. First, toy marketer’s descriptions of three IoToys through websites and digital and printed marketing materials were collected. Second, a semi-structured survey for preschool teachers was administered, and third, a semi-structured survey for the parents of the preschool children was conducted. The toys under investigation fulfill the criteria of IoToys: they are “smart,” and their connectivity usually occurs through mobile devices. In some cases, smart toys contain their own computers (e.g., the CogniToys Dino and Fisher-Price’s Smart Toy Bear.) The toys were chosen based on their age appropriateness, gender-neutrality as character types of toys, and their availability on Amazon US (in August 2017). Moreover, one reason for selecting these IoToys was the awards they have received. For example, the CogniToys Dino has received the Silver Honor 2016 Parent’s Choice Award. To analyze the research materials, a content analysis was

performed. The goal of a content analysis is to provide knowledge and understanding of the phenomenon under study [20]. It provides the researcher with the possibility to conduct a close reading of the data through a systematic classification process of coding and identifying themes and patterns. In this process, researchers immerse themselves in the data to allow new insights to emerge [21]. The method is also described as inductive category development [22].

3 Previous Research: Mobile Toys/Toys That Mobilize Players

This section outlines some traces of development of the toys that have been designed in the recent past and that either move by themselves or mobilize their players. The categorization begins with the area of robot toys.

Toy Mobility – Robot Toys

In recent years, new learning standards and best practices for integrating technology into early childhood education have gained increasing interest [23]. To teach technology and engineering to young children in a developmentally appropriate way, robotics and computer programming initiatives have grown in popularity among early education researchers and educators [24]. Since 1998, the Department of Electrical Engineering and Computer Engineering of the Université de Sherbrooke have been designing a variety of mobile robotic toys with the goal of using them as pedagogical tools for children suffering from autism or other developmental disorders [25–27]. Previous research has shown that the field of robotics holds special potential for early education by facilitating both cognitive and social development [28]. For example, recent research suggests that children as young as four years old can successfully build and program simple robotics projects while learning a range of engineering and robotics concepts in the process [29, 30]. In a study by Billard, which focuses on the novel application of Robota as a laboratory platform in an introductory class to robotics taught at the undergraduate level, they used a set of 10 Robota robots. Robota is a doll-shaped robot toy, which was developed with the goal of investigating social skills, and it is a human-shaped robot that children can have social interactions with [31]. In this study, the class of students learned to program the robot's micro-controller and the robot-PC interface using both vision and speech synthesizing. The lab session's aim was to develop games suitable for normal and disabled children [32].

Using coding toys and computer programming in early education has been claimed to support *cognitive* and *social* development. For example, studies with the text-based language *Logo* have shown that computer programming can help young children with a variety of cognitive skills, including number sense, language skills, and visual memory [33]. It is notable that many robotics activities do not involve sitting alone in front of a computer; rather, they encourage physically *mobile play*: robotic manipulatives allow children to develop fine motor skills and hand-eye coordination while also engaging in collaboration and teamwork [33]—a finding that the current case study with the Dash toy robot also supports.

Moreover, robots have been used as a part of playful learning through facilitation of *movement*. For example, Recio et al. used the Nao robot to assist a physiotherapist in modeling the movements performed during physiotherapist interventions. The study showed that they programmed the Nao robot so that it could perform nine exercises that were previously specified by the physiotherapists of the assisted living facility. The workshop consisted of one-to-one sessions with 13 participants. The results showed that Nao's movements seemed to be a highly relevant aspect to take into consideration when observing playful interactions with it [34].

Toy Mobility – Internet of Toys

Young children's growing fascination with interactive play is itself one of the clearest signs that the digital era is well underway for toys. As Negroponte claims, "it is almost genetic in its nature, in that each generation will become more digital than the preceding one" [1]. Children's media spaces and practices with the current Internet of Toys (IoToys) are digitally integrated to enable possibilities to learn through the mobility and flow of both content and data. The IoToys can interact in interesting ways, creating novel, appealing, and meaningful communication experiences and play situations using speech, sounds, visual cues, and movements; however, there is a subcategory of the IoToys, which seemingly affords more movement of the players than others—coding toys. For example, when children learn a programming language with the IoToys, they are not "just learning code, they are coding to learn" [35, 36]. As a consequence of IoToys being fitted with sensors and connecting to networks, these toys gain "new skills (affordances) that are present in new forms of communication" [37]. For example, with coding features, toys like the Dash Robot make children move physically [14]. The IoToys can react to various physical inputs (sounds, images, touch, and movements), which means that they track children's behavior and measure the surrounding environments and can interact with children on a personalized basis (e.g., by replying to their queries). In this way, the IoToys have become "media that has not been mediated before" [Ibid.]. *Mobile playful learning*, then, can be understood by analyzing contemporary smart toys such as toys with coding features, and those that allow players to code the toys' actions. Coding toys are usually toys that players mobilize through programming. In this way, mobility includes considerable potential to be put into play. For instance, Dash encourages children to move physically, such as running after the toy.

There are three general properties of IoToys. IoToys are: (1) *pervasive*—a smart, connected toy may follow a child through every activity (mobility toys or toys mobilized for players); they are (2) *social*—social aspects, such as multiplayer aspects, with one-to-one (for example, Wonder Workshop's Dash), one-to-many (Wonder Workshop's Dash has a "Play by Wonder Workshop" website for more activities, challenges, and pro-tips for playing with Dash), and *promote many-to-many relations* (the Blockly app for Dash provides the opportunity to share one's own coding tips with the Dash robot with the community); and they are (3) *connected*—the IoToys may connect and communicate with other toys, devices, and services through networks. "Many adults might believe these connected toys turn children into virtual drives down the information highway and anticipate a future where popular culture will be accommodated to their own deepest desires" [38–40].

Toys That Mobilize the Player – Toy Tourism

Currently, 21st century advances have provided connected play for both adults and children of all ages [41]. Kafai and Fields propose that “connected playthings are in the digital playground of the twenty-first century” [42, p. 3]. Children are now using online and offline and local and global connections in dynamic ways and a child may even play synchronously in a virtual world with others in ways that distinguish the nexus between private and public worlds [43]. One example is *toy tourism*—the mobilizing of physical toys. Toy tourism can be categorized as free-form and open-ended and thus as a creative play practice and as a form of play structured to be goal-oriented and therefore game-like. Toy tourism is defined as follows: toy tourism occurs when toys travel either as companions of their owners or “single-handedly” as organized by toy travel agencies [44].

In addition, physical toys are also made mobile through games, such as Geocaching. Connected play can involve using the Internet while playing with the physical toy and moving the toy and can also occur in the context of toy tourism all over the world, such as with the Travel Bugs in the Geocaching game. Geocaching is a high-tech, worldwide treasure hunt game, where a player hides a cache for others to find. Geocaching is now available in more than 185 countries. Travel Bugs are metal dog tags typically attached to another object, such as a toy. Each Travel Bug has its own unique tracking number stapled on it and has its own “diary” (a website) that follows its movements. The idea is to move a toy through the geocaches, where people can take the toy and another player can take the toy and change it to another cache. The tracking number of Travel Bugs is used as proof by the user that they found the item, and it also doubles as a way for the user to locate the personal website of the Travel Bug. Players move the toys, and their movement is recorded on a website. The idea is that by picking up and dropping off Travel Bug Trackables and reporting their movements on the Travel Bugs individual website, one is mirroring the Travel Bug’s real-world adventures [45].

4 Results

4.1 Mobile Toys

In this exploratory study, the goal was to understand toy mobility with its related experiences of players that move beyond the existing genres of computer-based experiences, which already exist with digital games and virtual worlds. By focusing on toy mobility and toys that mobilize players, the authors demonstrated key findings of previous research.

The results of the study showed that toy mobility contributes to a broader culture of play and edutainment, in which robot toys and IoToys, including coding toys develop new ways of thinking about toy computing and new ways of thinking about mobile playful learning. These new smart toys, robot toys, coding toys, and IoToys increase the understanding of the mobility of toys and are capable of encouraging players to both learn and become physically mobile and in this way, potentially enhance well-being.

Our study highlights how a player can interact with a robot toy, and when an activity such as coding is unlimited, it can continue as long as the player is interested. Some of the key values of robot toys are tactile exploration, expressivity, acting to change the state of the robot, the expressive potentialities of the toy robot through the use of colors and visual patterns (for example, the CogniToy Dino has a light that represents the mood of the dinosaur-shaped toy robot), shape and movement (such as the coding of the Dash robot for moving along a certain trail and letting other children try to catch it), the sense of touch in combination with proximity, and the use of the remote control to modify a robot's behavior. These findings are similar to those related to the Roball toy [46], which is a spherical robot capable of navigating in all types of environments without becoming stuck in some places or falling to the side. The interaction with Roball occurs through vocal messages and movements patterns, such as spinning, shaking, or pushing. According to a case study of Michaud et al., the majority of children were trying to catch Roball, to grab it, or to touch the robot [46].

Furthermore, the results showed that coding toys [47] can be programmed to respond differently to situations. For example, the IoToys can also learn over time through users' actions and can change the ways they respond to the world, generating more sophisticated interactions and unpredictable situations that can help capture and retain the player's interest. Therefore, IoToys encourage players to move. For example, in our case study, one preschooler coded Dash, and others became physically mobile by running and making a tunnel with their legs and allowing Dash to move through the tunnel [14]. This shows how children become creative in finding their own way to play with IoToys, which also mobilizes players. Similar results were observed by Yu and Roque [48], and the researchers mention that the toy robot Dot (Dash's companion) can cause the robot Dash to move, light up, or play a sound. They found that children program instructions based on different conditions or events through three main approaches: (1) sensors are usually embedded in physical robots to sense different environmental conditions so that the robots can be programmed to react accordingly; (2) for physical robots without sensors, maps are usually provided for children to decide how the robots will react to different conditions on a map; and (3) for virtual platforms, a series of commands or special effects, such as animations, sounds, and light effects, could be triggered when sprites interact without other objects on the screen. For example, the Curlybots can be synchronized to run together and in ScratchJr, several sprites can be programmed to move, play, make sounds, and change sizes at the same time [48].

Players can control the motion of several coding toys simultaneously or can program the motions, lights, and sound effects at the same time. For example, with the Dash and Dot toy robots employed in our own research, players can program the toys to interact with each other.

As the results have shown, toy computing is a recently developing concept in which traditional toys enter a new area of computer research using service computing technologies [49]. Therefore, in this context, a toy is a physical embodiment artifact that acts as a child's user interface for toy computing services in a cloud. The IoToys as an example of contemporary mobile toys can also capture a player's physical activity state (e.g., walking, running, etc.) and store personalized information (e.g., location, activity pattern, etc.) through a camera, microphone, Global Positioning System (GPS), and various sensors, such as facial recognition or sound detection [50].

4.2 Toys That Mobilize Players

According to the findings on our study on toys that mobilize players, following the development of historical automata, or self-moving toys, the commercial toy industry has developed a multitude of phygital play objects that have been given affordances related to sound, light, and movement. Many of these toys represent character toys, or toys with a face, such as dolls, action figures, or soft toys, with an emphasis on the idea of potential anthropomorphization. “The long-established fascination with automata and the need to animate anthropomorphic figures are still current,” Deborah Jaffé, author of *The History of Toys*, wrote in 2006 [11].

As shown, a player does not always keep his/her toys in the intimacy of the playroom in one’s living spaces. The results of this study showed that non-digital (character) toys, such as dolls, action figures or plushies can also mobilize players by making them geographically mobile by traveling with their toys, or setting them out to travel. *Toy tourism* may be divided into the two categories of either traveling with toys, or travel agencies moving others’ toys all over the world and playing the Geocaching game, where players move toys (Travel Bugs) from certain caches to other caches. In Geocaching, the main idea is to become mobile with the toy all over the world. This means that the toy is moving with different players, in this case while playing the geocaching game. For example, the Dr. Geocacher Travel Bug has moved 4041 km [51].

Outside of Geocaching, toys mobilize players by motivating them to more creative play than game-like competitions. One important facet of this activity is (visual) storytelling focusing on the toys travels. According to Robinson (2014), a traveling toy may have its own website or blog, a Twitter following, or a Facebook page. Travel agents for toys have appeared in the past decade as well as hosts who take the toy traveling and posts images – and the toy – back to the owner. Robinson’s study indicated that the Flickr photo service has 139 specific groups identified by Traveling Toys parameters, and the highest number of images uploaded and shared was 31,000 Traveling Toys (based on her survey in 2012) [52]. In this way, toys urge their players to travel and bring the toys along on the journeys. Similar play patterns can be detected among adult toy players [53, p. 285]. According to the findings, in order for toys to become mobile/mobilize, the traveling toys must be made *portable*: mobility in terms of toy play behavior relates to the affordances of toys. For example, toy company Kenner made its Star Wars figures pocket-sized so they could be carried around and played with wherever the player might choose [54, p. 213]. Since the beginnings of these massively popular action figures, many character toys are designed with mobility in mind. In other words, the players should be able to travel with their character toys, and in this way, expand the play environment beyond the ordinary living space of their players.

4.3 Classification of Mobilizing Technologies: The Toy Mobility Continuum

The focus of this section is the proposition of a structured classification of the identified mechanical, phygital, and digital realities of mobilizing technologies in relation to toys. This classification summarizes the main differences between these realities according to the two dimensions and is applicable to the design of mobility in toys. It also allows for

comparisons between reality and toy technologies. There are three areas (mechanical, phygital and digital toy mobility and analogue/digital toy tourism), which are referred to as “dimensions,” that relate either to a correlation of interactive toys that mobilize the player or a player traveling with toys.

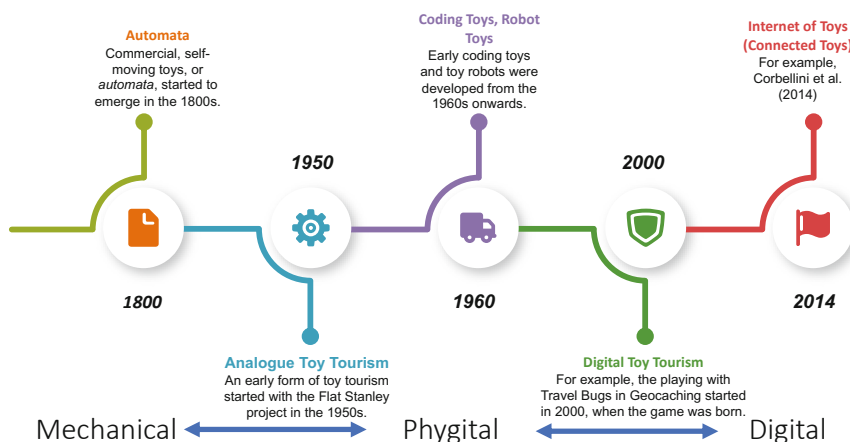


Fig. 2. Toy Mobility Continuum—a classification of toy-based mobilizing technologies.

Based on the findings, the Toy Mobility Continuum is proposed (see Fig. 2), which suggests a timeline of the development of phygital playful technology with mobilizing tendencies: toys that move by themselves or that mobilize their players. On one end of this timeline are the mechanical automata, which demonstrate toy inventors’, designers’ and artists’ desire to create self-moving toys. On the other end of continuum are the contemporary, connected toys, which represent both physical play objects as well as digitally connected playthings. Both the self-moving toys and the connected toys of the present have a relation to phygitality, one particularly due to their mechanics and the other due to their digitality.

5 Discussion

This exploratory paper has focused on the technological development of the toy medium with an interest in toys’ capacity to mobilize *homo ludens*, the playing human. By conducting an extensive literary review on the history and the present of mobile play objects, the way toys have developed from early moving and mechanical automata to playthings that move by themselves through in-built computerized components have been demonstrated. The interest of the paper was two-fold: By analyzing the historical trajectory of mobile toys, the authors highlighted the role of both toys and their players as participants in technologically mediated, phygital play. This hybrid form of playing combines the physicality of playthings with both mechanical and digital features.

Second, through a mixed methods approach, the findings of the literary review have been synthesized with findings of the authors' earlier empirical research conducted with preschool-aged children and their teachers with different types of contemporary interactive toys; toy robots and IoT toys.

The results of the study have illustrated how toys, character toys in particular, have transformed from entertaining, self-moving spectacles to educational machines that mobilize the player both physically and geographically. Based on the results of the study, the literary review, and previous empirical research, the *Toy Mobility Continuum* that visualizes the development of phygital playful technology and the types of toys that afford mobility in play of the past, present, and future has been proposed.

The limitation of the study is its conceptual approach—the authors based the findings of the paper on earlier studies and their own previous research as other studies in the field are limited. For example, there is no quantitative analysis of similar studies. Moreover, this study synthesized several case studies conducted earlier, but they are limited to Finnish participants only. However, although the present study only represents a firsthand approach in understanding the concept of toy mobility, it also provides an important contribution to understanding the ways that toys are a medium that mobilize in many ways. On the one hand, toys are given mobile affordances by their designers and makers. On the other hand, they are made mobile through the activities of the players, or make their players mobile, even by urging them to travel with their toys. Following these leads, our aim is that the tentative study presented in the paper at hand, will inspire ongoing and future studies of toy mobility and even generate new ideas regarding, for example, how toy mobility can be utilized as a starting point for innovating new products and services not only within the industries of play but also within the larger context of the experience economy interested in various forms of physical, cognitive, and social well-being, such as in education, and the health and tourism industries.

6 Conclusions and Future Studies

The main contribution of this study is that it highlights the mobility aspect of toys in many ways. Based on the findings, it was concluded that phygital, playful technologies, such as non-digital character toys, robot toys, or connected IoT toys, have the capacity to influence human well-being in various dimensions. By making players mobile, toys as a medium that invite engagement with phygital playful technologies have the potential to enhance physical, cognitive, and social well-being. Therefore, other researchers in the field of toys, both traditional and technologically-enhanced, are able to develop their understanding of toys' capacities in relation to mobility, mostly their mobilizing tendencies, which can be employed much more in the design of future toy-to-human interactions. More than before, toys can be used to increase human well-being by designing specific affordances into toys that result in well-being effects, such as physical (through exercise and geographical movement), cognitive (through educational benefits), and social (through social object play by playing together) well-being.

The novelty value of the paper is in presenting the Toy Mobility Continuum that emerged from the mixed method approach combining an extensive literature review with empirical research on contemporary toy play cultures with children, mature players, and transgenerational toy audiences. These observations would not have been possible to make through theoretical research only but required a creative approach: a synthesis of earlier work, both theoretical and empirical. Although extensive and unique in its research design, the limitation of this study is that it focuses on *character toys*, meaning dolls, action figures, and soft toys that in some cases have been turned into social robots. Therefore, a multitude of mobile toys and toys with mobilizing potential, such as toy vehicles were excluded in the analysis. Nevertheless, the study presented in this paper functions as an example of a firsthand approach and start for a scholarly dialogue on the mobilizing tendency of toys. To conclude, the main contribution of the paper is once more accentuated. By conducting this study, the way the pure physicality of the mechanically mobile toy has transformed into phygitality with a mobilizing capacity has been highlighted. It is assumed that the role of digital, social, and mobile technologies in particular will impact future toy design in unforeseeable ways. The future of toy play will involve an upswing in the integration of advanced, visual, and augmented technologies. For instance, the ubiquity of technology will mean more physical but screenless playthings with “hidden technologies” that through their phygitality continue to mobilize *homo ludens*, the playing human, in many more ways. Thus, to conclude, it is predicted that future toys will continue to influence well-being in more ways than before. One area of this development focuses on the physical mobility aspect of interactions with toys.

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